04 BESS Dispatch Simulation

Simulated 1MW/2MWh battery · optimal DAM cycle · gross revenue tracking

THE MODEL — KEEP IT SIMPLE AND CONSISTENT

Charge window	Find the 4 consecutive lowest-price half-hours. Charge cost = avg price × 2 MWh. Usually overnight or midday high-wind window.
Discharge window	Find the 4 consecutive highest-price half-hours after charging. Revenue = avg price × 2 MWh. Usually morning or evening peak.
Gross spread	Discharge revenue minus charge cost. This is daily gross profit before efficiency (assume 85% round-trip), network charges, and capacity costs.
Running total	Accumulate daily gross spread. Publish month-to-date and cumulative totals. Readers follow the number — it becomes a series.

ANGLES — ROTATE ONE PER POST

Good day / bad day	Was today a good day to be a BESS? High spread = yes. Low spread = no. What market condition drove it — wind, demand, season?
Wind correlation	High-wind days suppress price and shrink the spread. Quantify: every 10% increase in avg wind penetration costs approx €[X] in daily spread.
Cannibalisation paradox	The same wind that hurts renewable revenue creates the volatility that makes BESS valuable. Dual-peak high-wind days are the sweet spot.
Forecast vs perfect foresight	The model uses perfect DAM knowledge. A real operator uses forecasts. On a volatile day, how much worse would a forecast-based dispatch have done?
Seasonal trend	Is May a better month than April for BESS revenue? Plot month-to-date running totals side by side as data accumulates.
Stack value	DAM arbitrage is only one revenue stream. Ancillary services (DS3, FFR) sit on top. Acknowledge the stack — your number is a floor, not a ceiling.

HOOK SENTENCE FORMULA

Revenue day	*"A simulated 1MW BESS cycling today's DAM prices would have earned €[X] gross — charge at €[low] overnight, discharge into the [morning/evening] peak at €[high]."*
Poor day	*"Today's €[X] spread is the [weakest/strongest] this month. [High wind / low demand] compressed the intraday range to just €[Y]/MWh."*
Running series	*"Running month-to-date gross: €[X]. [Month] is tracking [above/below] April's €[Y] — and there are [N] trading days left."*

WHY THIS MATTERS FOR SALARY

Battery storage is the fastest-growing segment in European energy markets. Every developer, operator, trader, and investor is hiring analysts who understand dispatch economics. Publishing a running simulated BESS portfolio from real market data is rare, credible, and directly transferable to roles at storage developers, trading houses, and flexibility operators paying €70k–€110k+. This is your highest-value career post.

SALARY SIGNAL KEYWORDS TO USE

Dispatch optimisation	**Revenue stacking**	**Round-trip efficiency**	**Arbitrage spread**	**DS3 / ancillary services**